HW1

Ido Shimshi

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1. Normal Equations

Proof. At a minimizer, the gradient of $\mathcal{L}$ with respect to $\boldsymbol{\theta}$ is zero. Using $h_{\boldsymbol{\theta}}(\boldsymbol{x}_i) = \boldsymbol{\theta}^\mathsf{T}\boldsymbol{x}_i$,

$$\begin{aligned}\nabla_{\boldsymbol{\theta}}\mathcal{L} &= \nabla_{\boldsymbol{\theta}}\left[\frac{1}{2M}\sum_{i=1}^{M}(\boldsymbol{\theta}^\mathsf{T}\boldsymbol{x}_i - y_i)^2\right] \\ &= \frac{1}{M}\sum_{i=1}^{M}(\boldsymbol{\theta}^\mathsf{T}\boldsymbol{x}_i - y_i)\boldsymbol{x}_i \\ &= \frac{1}{M}\boldsymbol{X}^\mathsf{T}(\boldsymbol{X}\boldsymbol{\theta} - \boldsymbol{y}).\end{aligned}$$

Here, $\nabla_{\boldsymbol{\theta}}(\boldsymbol{\theta}^\mathsf{T}\boldsymbol{x}_i - y_i) = \boldsymbol{x}_i$, since $\boldsymbol{x}_i$ and y_i are fixed with respect to $\boldsymbol{\theta}$. Therefore,

$$\mathbf{0} = \frac{1}{M}\boldsymbol{X}^\mathsf{T}(\boldsymbol{X}\boldsymbol{\theta} - \boldsymbol{y}) \quad \Longrightarrow \quad \boldsymbol{X}^\mathsf{T}\boldsymbol{X}\boldsymbol{\theta} = \boldsymbol{X}^\mathsf{T}\boldsymbol{y}.$$

□

2. Unique Solution

Proof. We prove both directions.

If $\boldsymbol{X}$ has full column rank, the minimizer is unique. Full column rank means that $\boldsymbol{X}\boldsymbol{v} = \mathbf{0}$ only when $\boldsymbol{v} = \mathbf{0}$. Thus, for every nonzero $\boldsymbol{v}$,

$$\boldsymbol{v}^\mathsf{T}\boldsymbol{X}^\mathsf{T}\boldsymbol{X}\boldsymbol{v} = (\boldsymbol{X}\boldsymbol{v})^\mathsf{T}(\boldsymbol{X}\boldsymbol{v}) = \|\boldsymbol{X}\boldsymbol{v}\|^2 > 0,$$

so $\boldsymbol{X}^\mathsf{T}\boldsymbol{X}$ is invertible. Let

$$\boldsymbol{\theta}^* = (\boldsymbol{X}^\mathsf{T}\boldsymbol{X})^{-1}\boldsymbol{X}^\mathsf{T}\boldsymbol{y}.$$

Then $\boldsymbol{X}^\mathsf{T}(\boldsymbol{X}\boldsymbol{\theta}^* - \boldsymbol{y}) = \mathbf{0}$, so $\boldsymbol{\theta}^*$ satisfies the normal equations. For any $\boldsymbol{\theta}$, write $\boldsymbol{d} = \boldsymbol{\theta} - \boldsymbol{\theta}^*$. Using $\|\boldsymbol{a} + \boldsymbol{b}\|^2 = \|\boldsymbol{a}\|^2 + 2\boldsymbol{a}^\mathsf{T}\boldsymbol{b} + \|\boldsymbol{b}\|^2$,

$$\begin{aligned}\mathcal{L}(\boldsymbol{\theta}) - \mathcal{L}(\boldsymbol{\theta}^*) &= \frac{1}{2M}\left(\|\boldsymbol{X}\boldsymbol{\theta}^* - \boldsymbol{y} + \boldsymbol{X}\boldsymbol{d}\|^2 - \|\boldsymbol{X}\boldsymbol{\theta}^* - \boldsymbol{y}\|^2\right) \\ &= \frac{1}{2M}\left(\|\boldsymbol{X}\boldsymbol{\theta}^* - \boldsymbol{y}\|^2 + 2(\boldsymbol{X}\boldsymbol{\theta}^* - \boldsymbol{y})^\mathsf{T}\boldsymbol{X}\boldsymbol{d} + \|\boldsymbol{X}\boldsymbol{d}\|^2 - \|\boldsymbol{X}\boldsymbol{\theta}^* - \boldsymbol{y}\|^2\right) \\ &= \frac{1}{2M}\left(2(\boldsymbol{X}\boldsymbol{\theta}^* - \boldsymbol{y})^\mathsf{T}\boldsymbol{X}\boldsymbol{d} + \|\boldsymbol{X}\boldsymbol{d}\|^2\right) \\ &= \frac{1}{2M}\|\boldsymbol{X}\boldsymbol{d}\|^2 \geq 0.\end{aligned}$$

The remaining cross term is zero because

$$(\boldsymbol{X}\boldsymbol{\theta}^* - \boldsymbol{y})^\mathsf{T}\boldsymbol{X}\boldsymbol{d} = \left(\boldsymbol{X}^\mathsf{T}(\boldsymbol{X}\boldsymbol{\theta}^* - \boldsymbol{y})\right)^\mathsf{T}\boldsymbol{d} = 0,$$

Equality holds only if $\boldsymbol{X}\boldsymbol{d} = \boldsymbol{0}$, which by full column rank implies $\boldsymbol{d} = \boldsymbol{0}$. Therefore $\boldsymbol{\theta}^*$ is the unique minimizer.

If the minimizer is unique, $\boldsymbol{X}$ has full column rank. Suppose instead that $\boldsymbol{X}$ does not have full column rank. Then there is a nonzero vector $\boldsymbol{v}$ such that $\boldsymbol{X}\boldsymbol{v} = \boldsymbol{0}$. For a minimizer $\boldsymbol{\theta}^*$,

$$\mathcal{L}(\boldsymbol{\theta}^* + \boldsymbol{v}) = \frac{1}{2M}\|\boldsymbol{X}(\boldsymbol{\theta}^* + \boldsymbol{v}) - \boldsymbol{y}\|^2 = \frac{1}{2M}\|\boldsymbol{X}\boldsymbol{\theta}^* - \boldsymbol{y}\|^2 = \mathcal{L}(\boldsymbol{\theta}^*).$$

Since $\boldsymbol{v} \neq \boldsymbol{0}$, this gives a different minimizer, contradicting uniqueness. Hence $\boldsymbol{X}$ must have full column rank. □